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Diagnostic accuracy of traditional and deep learning methods for detecting depression based on speech features: a systematic review and meta-analysis

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Abstract

Background Depression diagnosis faces challenges of subjectivity and delay. Speech features offer potential objective biomarkers, but a systematic comparison of traditional machine learning (TML) and deep learning (DL) models is lacking.

Objective To evaluate and compare the diagnostic accuracy of TML and DL models for depression detection using speech features, and to examine subgroup effects across sample size, validation strategy, language, and diagnostic criteria.

Methods Following PRISMA guidelines, we systematically searched 9 databases (PubMed, Medline, Embase, PsycINFO, Scopus, IEEE, Cochrane, ACM Digital Library, and Web of Science) from inception to April 2025. Eligible studies included clinically diagnosed patients with depression and healthy controls, assessed using speech-based TML or DL models, and reporting sensitivity, specificity, or the area under the curve (AUC). Risk of bias was evaluated using the diagnostic Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2). Random-effects bivariate models pooled diagnostic performance, and heterogeneity, subgroup, and sensitivity analyses were conducted.

Results Twenty-five studies met the inclusion criteria (9 TML, 16 DL). TML models showed pooled sensitivity of 0.82 (95% CI: 0.74–0.88), specificity 0.83 (95% CI: 0.75–0.90), and AUC 0.89 (95% CI: 0.86–0.92). DL models achieved pooled sensitivity of 0.83 (95% CI: 0.77–0.88), specificity 0.86 (95% CI: 0.80–0.90), and AUC 0.91 (95% CI: 0.89–0.93). Subgroup analyses indicated that diagnostic performance varied by sample size, validation strategy, language, and diagnostic criteria.

Conclusion Both TML and DL models demonstrate good diagnostic accuracy in speech-based depression detection. The marginal but consistent superiority of DL models supports their potential use in secondary care settings for confirmatory diagnosis, while TML remains valuable for primary care screening..

Clinical trial number Not applicable.

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Keywords Depression, Speech features, Traditional machine learning, Deep learning, Diagnostic accuracy, Systematic review, Meta-analysis

Introduction

Depression is a major global public health challenge, affecting about 4% of the world's population or more than 300 million people. In 2021, around 730,000 people died by suicide, making it the third leading cause of death among those aged 15 to 29 years [1]. Patients often exhibit symptoms such as low mood, diminished interest, and difficulty concentrating, which not only severely affect their quality of life but may also lead to long-term disability or even suicidal behavior, imposing a heavy burden on individuals, families, and society. A research report stated that the socioeconomic burden of Major Depressive Disorder (MDD) in the United States was estimated to be $333.7 billion in 2019 [2]. Early and accurate identification of depression followed by timely intervention is crucial for reducing treatment complications, shortening the course of illness, and improving prognosis. However, current depression diagnosis faces multiple challenges.

The symptoms of depression are highly heterogeneous and nonspecific, often coexisting with other physical or mental illnesses, which leads to difficulties in differential diagnosis [3]. Self-reports of patients are significantly influenced by cultural background, cognitive abilities, and subjective willingness [4].Additionally, current diagnostic criteria require symptoms to persist for at least two weeks. Coupled with the insufficient use of standardized assessment tools and the lack of objective biomarkers, the average diagnostic delay lasts more than 2 years, with some patients even waiting up to 5 years [5–7]. Delayed diagnosis may not only trigger hippocampal atrophy and prefrontal dysfunction, reducing the effectiveness of antidepressant treatment [8]. but also exacerbate disturbances in neurotransmitter systems (such as serotonin and dopamine), leading to more irreversible biological changes [9].

The screening of traditional scales (such as the Patient Health Questionnaire, PHQ-9) has a false positive rate as high as 50% in primary care. Even with relatively accurate semi-structured interviews, there is still a significant risk of misdiagnosis due to nonspecific symptoms, patients' subjective concealment, and insufficient clinical experience of physicians [10–12]. Cultural differences further exacerbate the complexity of diagnosis. For example, Tamil patients tend to describe depression using religious metaphors [13], African Americans may express it through negative emotions and interpersonal conflicts [14], while immigrant patients often have difficulty accurately stating symptoms due to language barriers or cultural taboos (such as concepts of privacy) [15]. Such cultural variability also raises questions about whether speech-based acoustic features such as pitch variability or prosodic attenuation are valid across languages and cultural groups. Establishing the cross-cultural generalizability of speech biomarkers is therefore essential before clinical implementation.

In the face of these diagnostic dilemmas, researchers have been continuously exploring new diagnostic methods and biomarkers. In recent years, speech characteristics have gradually become a new direction for the objective diagnosis of depression. The speech of individuals with depression exhibits features such as slowed speech rate, prolonged pauses, and reduced fundamental frequency fluctuations [16, 17], which are closely associated with abnormal function of the limbic system [16]. By integrating multi-dimensional bio-acoustic indicators such as sound source characteristics, spectral features, prosodic patterns, and formant parameters, the classification performance of depression detection models can be significan tly improved [18, 19]. Specifically, there are significant differences in acoustic parameter distribution, time-domain dynamic characteristics, and spectral energy patterns between the speech of depressed patients and healthy individuals [20]. These differences can serve as quantitative indicators to provide technical support for grading symptom severity and evaluating treatment efficacy [21].

Compared with the single assessment of traditional scales, speech data can be collected remotely and frequently through smartphone applications. For example, the analysis of speech fragments from human-computer conversations or clinical interviews can achieve a screening accuracy rate of 93.3%, and model performance improves with the increase in audio duration and quantity [22]. Voice features collected by smartphones (such as fundamental frequency, pitch, etc.) could track antidepressant treatment response in real time. For example, the semitone fluctuation of fundamental frequency (F0) can reflect changes in emotional states, while energy distribution features can dynamically monitor symptom progression [23]. This continuous monitoring capability compensates for the lag in traditional methods and provides an objective basis for personalized treatment.

Machine learning (ML) technologies have provided powerful tools for the analysis of speech characteristics. ML encompasses Traditional Machine Learning (TML) (such as Support Vector Machines (SVM) and random forests) and Deep Learning (DL) models (such as convolutional neural networks and Transformer). By extracting acoustic features like Mel-frequency cepstral coefficients

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(MFCCs), zero-crossing rate (ZCR), and harmonic-to-noise ratio (HNR), ML bypasses traditional text analysis to directly capture acoustic signatures associated with depression [24, 25]. This approach offers significant advantages in multilingual and cross-cultural contexts, effectively avoiding the diagnostic biases caused by linguistic differences or cultural expression habits.

Machine learning and deep learning have shown promise across various medical domains, such as in predicting COVID-19 hospitalization where optimized algorithms improved risk stratification and clinical decision-making [26]. Similar studies further demonstrate how advanced ML frameworks enhance diagnostic accuracy in diverse conditions [27]. In the context of psychiatry, AI models for speech-based depression diagnosis have also shown encouraging accuracy in laboratory settings; however, their real-world generalizability remains limited by small sample sizes, non-representative datasets, and training biases [28, 29]. A recent review by Liu et al. (2024) [30] systematically quantified these challenges, noting that 68% of studies relied on English datasets and only one-third reported sufficient diagnostic indices for meta-analysis, underscoring how language imbalance and inadequate external validation restrict robustness and clinical applicability of current models.

In primary healthcare institutions with limited hardware resources, the application of deep learning (DL) models, particularly convolutional neural networks (CNNs), is constrained by their substantial requirements for Graphics Processing Unit (GPU) capacity and large-scale data processing [31]. At the same time, clinicians have expressed concerns about the "black box" nature of AI. A survey reported that 39.5% of oncologists were worried about the opacity of AI decision-making [32], and other studies have similarly noted that the lack of interpretability may reduce trust and hinder broader clinical adoption [33, 34]. This had limited their wide application in clinical practice to a certain extent.

Even though several studies have developed speech feature-based TML and DL models for depression prediction, most focus on a single model type and lack a systematic comparison of their diagnostic accuracy. This gap limits a clear understanding of the relative diagnostic performance of the two model types, including the area under the curve (AUC), sensitivity, and specificity. It also makes it difficult to determine how key methodological factors, such as sample size, validation strategy, language, and speech task type, influence model performance.

Objectives: Therefore, the present systematic review and meta-analysis was designed to (1) compare the pooled diagnostic accuracy of TML and DL models in speech-based depression detection, and (2) examine how subgroup factors including sample size, validation strategies, diagnostic criteria, and language affect model performance. We hypothesized that DL models will outperform TML models in speech-based depression detection, and that subgroup factors (e.g., larger sample size, rigorous validation strategies, consistent diagnostic criteria, and English-language datasets) will be associated with better model performance.

Methods

The meta-analysis took the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guideline [35] as the reference. The study was registered in the PROSPERO (CRD42024595530). The registration date was registered 29/9/2024.

Search strategy

The systematic search of PubMed, Medline, Embase, PsycINFO, Scopus, IEEE, Cochrane and Web of Science databases for relevant literature published from inception to 8 April 2025 was performed. To ensure the inclusion of the most recent and comprehensive evidence, an updated search was performed on October 10, 2025, during which the ACM Digital Library was newly added to the search sources. The search strategy used a combination of subject terms and free words, with keywords including but not limited to: depression, depressive disorder, deep learning, machine learning, artificial intelligence, automat, sound, speech, voice, acoustic, audio, vessel, vocal, pitch, prosody, etc. and their combinations. The complete retrieval strategy taking PubMed as an example is shown in Table 1 of the Supplementary File (including Boolean operators and field tags).

Selection criteria

Inclusion criteria

(1) Population (P): Clinically diagnosed depression patients, with control groups being healthy individuals or other mentally ill patients explicitly excluded of depression (e.g., anxiety disorder, schizophrenia, etc.). (2) Index Test (I): Machine learning models based on speech features, including TML and DL. (3) Comparator (C): The gold standard is clinical interview combined with the *Diagnostic and Statistical Manual of Mental Disorders* (DSM) or the International Classification of Diseases (ICD) diagnostic criteria, or comparison with healthy controls using standardized scales (e.g., PHQ-9 [36], the Hamilton Depression Scale (HAMD) [37]). (4) Outcomes (O): At least one diagnostic performance index must be reported (sensitivity, specificity and AUC), or raw data of confusion matrix (true positive (TP), false positive (FP), true negative (TN), false negative (FN) must be provided. (5) Study Design (S): Cross-sectional studies, case-control studies, or cohort studies.

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Table 1 Subgroup analysis of TML and DL models

Model	Subgroups	Levels	Number	Sen (95%CI)	Spe (95%CI)	PLR (95%CI)	NLR (95%CI)	DOR (95%CI)	SROC (95%CI)	I^2
TML	Sample size	≤ 100	6	0.81 (0.74–0.87)	0.76 (0.67–0.83)	3.3 (2.4–4.6)	0.25 (0.17–0.35)	13 (7–25)	0.86 (0.82–0.88)	100%
		>100	5	0.82 (0.67–0.91)	0.87 (0.73–0.95)	6.5 (2.6–16.4)	0.21 (0.10–0.43)	32 (7-151)	0.91 (0.88–0.93)	68%
	Validation strategy	Grouped validation	6	0.85 (0.75–0.92)	0.89 (0.77–0.95)	7.6 (3.4–17.0)	0.16 (0.09–0.31)	46 (12–173)	0.93 (0.91–0.95)	68%
		Non-grouped validation	5	0.69 (0.66–0.71)	0.72 (0.70–0.75)	2.5 (2.3–2.7)	0.43 (0.40–0.47)	6 (5–7)	0.77 (0.73–0.80]	100%
	scales	DSM	6	0.78 (0.69–0.85)	0.71 (0.59–0.81)	2.7 (1.8–4.2)	0.31 (0.20–0.47)	9 (4–20)	0.82 (0.78–0.85)	0%
		Other scales	5	0.84 (0.70–0.92)	0.89 (0.79–0.95)	7.7 (3.6–16.6)	0.18 (0.09–0.36)	42 (11–162)	0.93 (0.90–0.95)	0%
	Speech task type	Interview	6	0.85 (0.73–0.92)	0.85 (0.72–0.92)	5.6 (2.6–12.1)	0.18 (0.08–0.37)	32 (7-140)	0.92 (0.89–0.94)	31%
		Reading	5	0.76 (0.67–0.84)	0.83 (0.71–0.91)	3.2 (2.2–4.7)	0.29 (0.19–0.43)	11 (6–22)	0.83 (0.80–0.86)	68%
DL	Sample size	≤ 100	6	0.75 (0.61–0.85)	0.85 (0.77–0.90)	4.5 (2.6–7.7)	0.29 (0.20–0.40)	16 (8–31)	0.86 (0.82–0.88)	0%
		100–500	5	0.86 (0.77–0.92)	0.80 (0.74–0.85)	4.2 (3.3–5.4)	0.18 (0.11–0.29)	24 (14–41)	0.87 (0.84–0.90)	82%
		≥ 500	9	0.84 (0.74–0.91)	0.88 (0.78–0.94)	7.2 (3.4–15.3)	0.18 (0.10–0.33)	40 (11–144)	0.93 (0.90–0.95)	100%
	Validation strategy	Grouped validation	4	0.88 (0.81–0.93)	0.83 (0.74–0.89)	5.2 (3.5–7.8)	0.18 (0.10–0.33)	37 (25–56)	0.92 (0.90–0.94)	83%
		Non-grouped validation	16	0.82 (0.74–0.87)	0.87 (0.80–0.92)	6.2 (3.7–10.1)	0.21 (0.14–0.32)	29 (12–69)	0.91 (0.88–0.93)	100%
	Language	Chinese	7	0.87 (0.85–0.88)	0.81 (0.71–0.89)	4.6 (2.9–7.4)	0.16 (0.14–0.19)	28 (16–50)	0.87 (0.84–0.90)	98%
		English	10	0.82 (0.70–0.90)	0.90 (0.83–0.95)	8.3 (4.2–16.3)	0.20 (0.11–0.36)	41 (12–138)	0.93 (0.90–0.95)	100%
		Other	3	0.57–0.74	0.66–0.88	-	-	-	-	
	scales	DSM	5	0.86 (0.83–0.89)	0.86 (0.77–0.91)	6.1 (3.7–9.8)	0.16 (0.13–0.20)	38 (21–70)	0.87 (0.84–0.90)	88%
		Other scales	15	0.81 (0.73–0.88)	0.86 (0.79–0.92)	5.9 (3.5–10.0)	0.21 (0.14–0.33)	28 (11–68)	0.91 (0.88–0.93)	100%
	Speech task type	Interview	15	0.83 (0.76–0.88)	0.88 (0.82–0.93)	7.0 (4.2–11.5)	0.19 (0.13–0.29)	36 (16–85)	0.92 (0.89–0.94)	100%
		Reading	5	0.84 (0.70–0.92)	0.77(0.68–0.85)	3.7 (2.5–5.6)	0.21 (0.10–0.42)	18 (7–50)	0.86 (0.83–0.89)	85%
	Deep Learning Architecture	Traditional	10	0.86 (0.80–0.90)	0.90 (0.84–0.94)	8.2 (4.9–13.6)	0.16(0.11–0.24)	52(22–123)	0.94 (0.91–0.96)	98%
		Advanced CNNs	10	0.81(0.70–0.89)	0.82(0.71–0.90)	4.6 (2.6–7.9)	0.23 (0.14–0.39)	20(7–53)	0.89 (0.85–0.91)	100%

Exclusion criteria

(1) Studies for which the full text was unavailable. (2) Animal studies, reviews, conference abstracts, and case reports. (3) Multimodal studies were excluded. Although such approaches may enhance diagnostic performance, their results cannot be attributed solely to speech features. To ensure methodological consistency and comparability of pooled estimates, this review focused only on speech-based models. (4) Studies that relied solely on text features (Natural Language Processing, NLP) or non-speech physiological signals (e.g., electroencephalogram - EEG, electrocardiogram - ECG, functional magnetic resonance imaging - fMRI) for depression detection. (5) Studies that did not report raw data, i.e., those that did not provide results of diagnostic tests (TP, FP, TN and FN) or related derived indicators. (6) Studies that did not use machine learning algorithms. (7) Studies without a clear separation of training and testing sets, or without

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an independent test set, nested k-fold ($k \geq 5$), or subject-level cross-validation (e.g., leave-one-subject-out). Random utterance-level splits, single random splits, or any method with potential train–test overlap were excluded.

Study selection and data extraction

The retrieved literature was imported into EndNote software and duplicates were first eliminated. Two researchers independently and blindly screened titles, abstracts, and full texts against the pre-established criteria. Discrepancies were resolved through discussion or consultation with a third reviewer.

Extracted information included: authors, year, country, study design, patient source, diagnostic criteria, language, speech features, feature extraction software, model type, sample size, diagnostic test results (TP, FP, TN, FN), and validation strategy. For diagnostic test data, two reviewers independently reconstructed 2×2 tables when needed. If discrepancies arose (e.g., inconsistent reporting of TP/FP/TN/FN across text, tables, or figures), they were resolved by applying predefined rules (e.g., prioritizing raw counts over derived indices) and verified by consensus with a third reviewer.

Risk of bias assessment

The quality of the included literature was assessed by two investigators using the diagnostic Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) [38] recommended by the National Institute for Health and Clinical Excellence. In case of disagreement between the two researchers, a third researcher was brought in to reach a final consensus. Risk of bias was assessed with QUADAS-2, adapted for AI studies by adding specific signaling questions. For Patient Selection, we asked whether participants reflected the intended clinical population and whether speech samples were independent across subjects. For the Index Test, we evaluated whether model development and validation were strictly separated, whether preprocessing and feature selection were confined to training data, and whether outcome information was unavailable during training. For the Reference Standard, we checked if diagnoses were based on validated clinical criteria (DSM/ICD or standardized scales). For Flow and Timing, we assessed whether the time interval between speech recording and reference diagnosis was short (≤ 2 weeks) and whether all patients were included in the analysis. Additional AI-specific questions included: risk of train–test leakage, appropriateness of unit of analysis (subject vs. utterance level), handling of class imbalance, and reporting of missing data. The Risk of bias and applicability concerns summary can be found in Supplementary File (Fig. 1).

Data analysis

All analyses were performed in Stata 18 using midas/metandi routines. From each eligible study, two reviewers independently extracted TP, FP, TN, FN; discrepancies were resolved by predefined rules (e.g., prioritizing raw counts over derived indices) and consensus with a third reviewer. Forest plots of sensitivity and specificity and summary ROC curves were generated.

We used the bivariate random-effects model (Reitsma) to jointly estimate pooled sensitivity and specificity, accounting for potential threshold effects. Threshold effects were evaluated using Spearman correlation between logit-transformed sensitivity and specificity. When threshold effects were suspected, we additionally fitted a hierarchical summary receiver operating characteristic (HSROC) model as a sensitivity analysis. The bivariate random-effects model was chosen as the primary analytic framework because it directly provides pooled estimates of sensitivity and specificity while accounting for their correlation. This approach is mathematically equivalent to the HSROC model under most conditions, but offers more clinically interpretable summary points, making it widely recommended in diagnostic accuracy meta-analyses.

We summarized between-study heterogeneity using I^2 (for sensitivity and specificity), between-study variance (τ^2), and 95% Confidence Interval (CI). We did not apply a rigid I^2 cutoff; instead, we interpreted heterogeneity in conjunction with τ^2 and 95%CI, and considering clinically and methodologically relevant differences among studies (e.g., population, diagnostic criteria, language, speech task, validation strategy).

We performed bivariate meta-regression to assess the impact of algorithm type (DL vs. TML). Pre-specified subgroup analyses included sample size (≤ 100 vs. > 100), validation strategy (independent test set vs. grouped cross-validation), and language. Exploratory analyses covered diagnostic criteria, speech task type, and DL architecture. For small subgroups ($n < 4$), we reported value ranges. Model robustness was evaluated via leave-one-out and extreme-size study exclusion, while publication bias was assessed using Deeks' funnel asymmetry test ($P > 0.05$ indicating no significant asymmetry).

Results

Literature screening process and results

The initial database search (April 2025) identified 4,428 records from eight databases. To enhance coverage, an updated search including the ACM Digital Library was conducted on October 10, 2025, yielding 216 new records. These additional studies were screened using the same inclusion and exclusion criteria and integrated into the overall PRISMA flow. This brought the total number of identified records to 4,644. All records were imported

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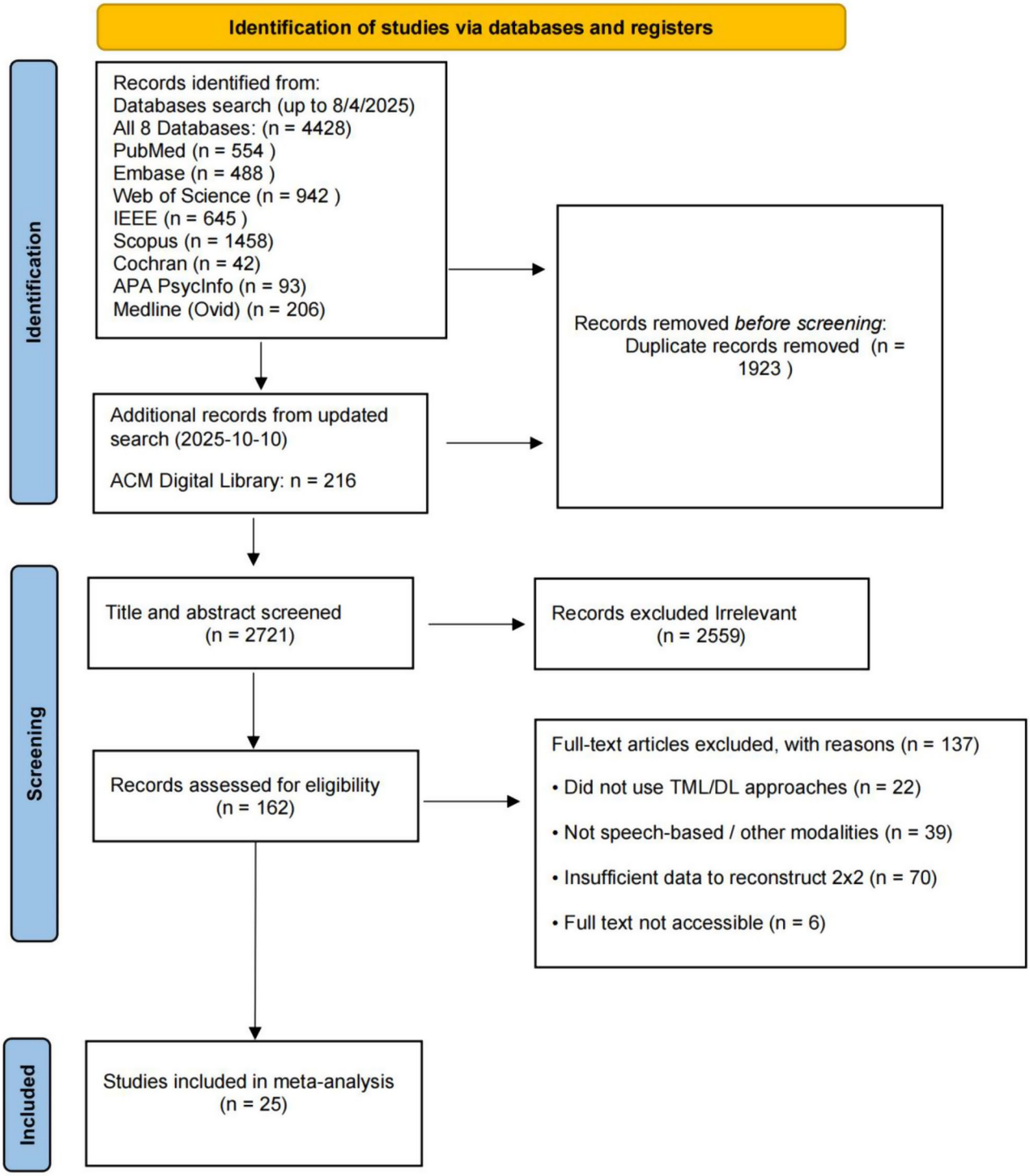


Fig. 1 PRISMA diagram of the study selection flow

into EndNote for deduplication, removing 1,923 duplicates and leaving 2,721 unique articles for screening. During title and abstract screening, 2,559 studies were excluded because they were reviews, conference reports, or meeting abstracts that did not meet the objectives or eligibility criteria of this study. A total of 162 full-text articles were reviewed in detail against the eligibility criteria.

After full-text review, 137 articles were excluded for the following reasons: ①22 articles investigated the role of speech signal processing in the detection of depression but did not use TML or DL algorithms; ②39 articles did

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not use speech samples to detect depression or used data in other formats (e.g., text, emotion, image, video, and electroencephalogram, etc.); ③70 studies did not report sufficient information to reconstruct a 2 × 2 table, such as reporting only AUC or accuracy without sensitivity and specificity, or providing precision/recall metrics without class counts (TP, FP, TN, FN). Some also reported results at the utterance level or as cross-validation averages, which prevented subject-level derivation. ④6 articles were excluded because they did not have access to the full text. Finally, 25 articles were included in this meta-analysis (Fig. 1).

Characteristics of the included studies

A total of 25 studies were included in this meta-analysis. The studies were primarily case-control or cross-sectional in design and were conducted across several countries, including Brazil, China, France, Germany, Hungary, Iran, Japan, Malaysia, Thailand, Turkey, and the USA.

Participants were recruited from a variety of settings, such as hospitals (e.g., Ulisses Pernambucano Hospital and Peking University Sixth Hospital), academic institutions (e.g., Saarbrücken University), and publicly available datasets (e.g., DAIC-WOZ, AVEC 2013/2014, MODMA). The sample sizes of the depression and HC groups ranged from 22 to 1650 and 11 to 1650, respectively. The diagnosis of depression was primarily based on the DSM-5/IV criteria. Standardized scales, including the PHQ-8/9, BDI-II, and the HAMD-17, were also used for assessment.

The languages analyzed in the included studies were diverse, covering English, Chinese (Mandarin), Portuguese, Hungarian, German, Japanese, Malay, Thai, and Turkish. Speech data were collected through two main task types: clinical interviews and read speech. A variety of speech features were extracted for analysis, including: Spectral features (e.g., MFCCs and spectrograms). Software tools such as GNU Octave, Praat, OpenSMILE, MATLAB, and Librosa were used for feature extraction.

The studies employed both TML models, such as SVM, RF, and LR, and DL models, including CNN, Transformers, and LSTM networks. To ensure robust performance evaluation, various validation strategies were implemented, such as k-fold cross-validation, LOOCV, and the use of independent test sets. Detailed characteristics of each included study are provided in Supplementary Table 2.

Risk of bias assessment for included studies

The methodological quality of the included studies was evaluated using the QUADAS-2 tool (Fig. 2). Overall, studies exhibited moderate risk of bias with variability across domains. Most studies were rated as low risk for Patient Selection because participants were recruited from well-defined clinical or public datasets; however, several used convenience samples or unbalanced case–control ratios. In the Index Test domain, about half of the studies were rated as high or unclear risk due to insufficient description of training testing separation and lack of blinding to reference standards during model development. The Reference Standard domain was generally low risk when DSM-based clinical interviews were applied, but unclear when only self-report questionnaires (e.g., PHQ-9, BDI-II) were used. For Flow and Timing, most studies achieved short assessment intervals (≤ 2 weeks) and complete inclusion, though several failed to specify timing or excluded participants post-hoc. Detailed item-level justifications are presented in Supplementary Fig. 1.

Meta-analysis results

The 9 TML studies were included, with a total of 11 TML models, involving 5882 speech samples. The meta-analysis showed the pooled sensitivity of 0.82 (95% CI: 0.74–0.88), pooled specificity of 0.83 (95% CI: 0.75–0.90), and AUC was 0.89 (95% CI: 0.86–0.92), PLR and NLR were 4.9 (95% CI: 3.0-8.2) and 0.22 (95% CI: 0.14–0.34), DOR was 22 (95% CI: 9.0–55.0). Figures 3 and 4 represents the pooled sensitivity and specificity forest plots and SROC cure of TML.

The 16 DL studies were included, with a total of 20 DL models, involving 701,063 speech samples. Similarly, we used the evaluation parameters of sensitivity, and specificity using a random-effects model, the pooled sensitivity 0.83 (95% CI: 0.77–0.88), pooled specificity was 0.86 (95% CI: 0.80–0.90), and AUC was 0.91 (95% CI: 0.89–0.93). PLR was 6.0 (95% CI: 4.0-8.9), NLR and DOR were

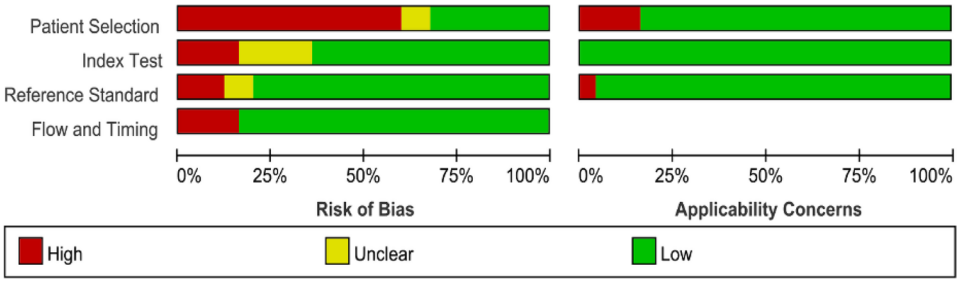


Fig. 2 Quality assessment results of included studies based on QUADAS-2 tool criteria

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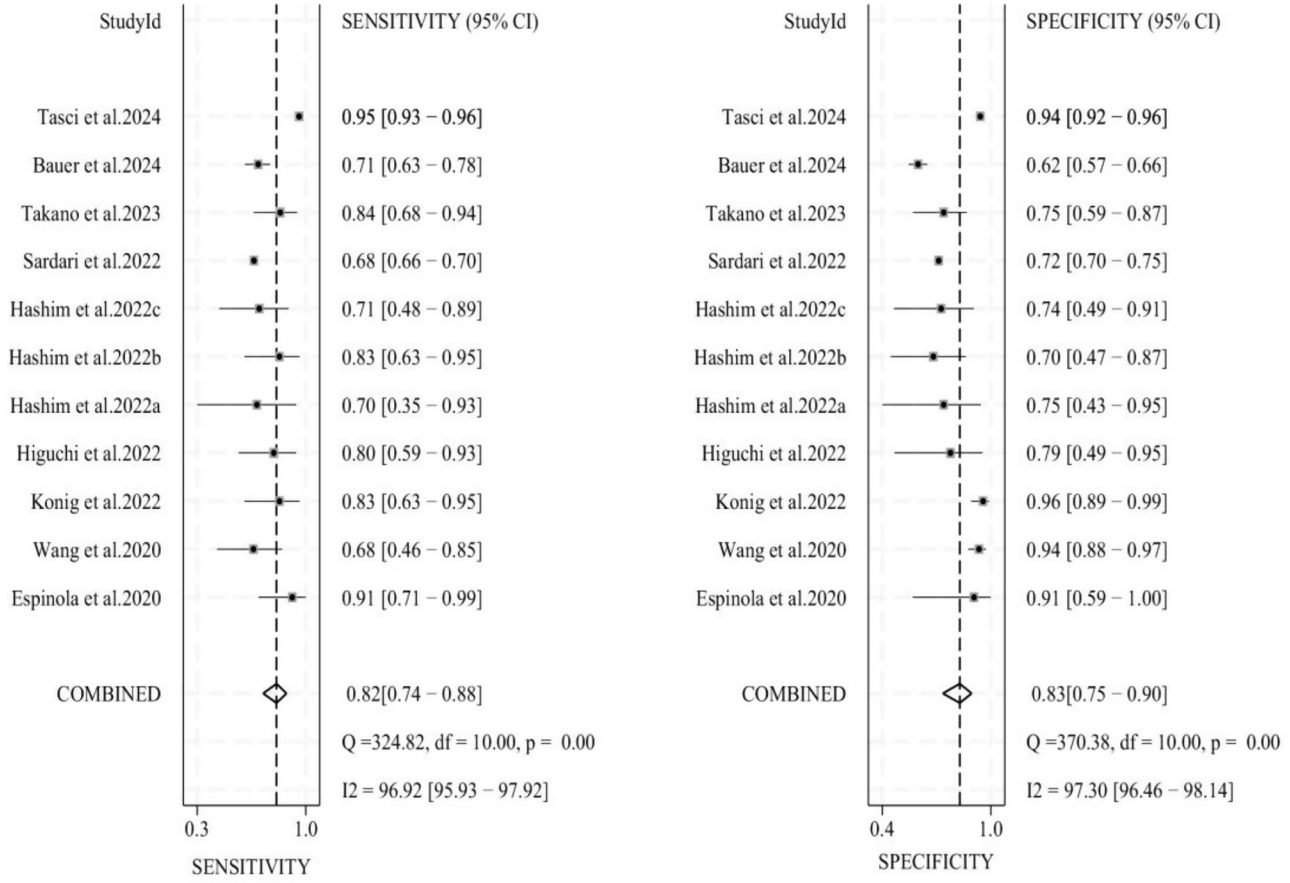


Fig. 3 The pooled sensitivity and specificity forest plots of TML. (B) SROC with prediction and confidence contours of TML

0.20 (95% CI: 0.14–0.28) and 31 (95% CI: 15–61). Figures 5 and 6 represents the forest plots of pooled sensitivity and specificity and SROC curve of DL.

Subgroup analysis

We further conducted a bivariate meta-regression with algorithm type (DL vs. TML) as a covariate. DL models showed slightly higher pooled sensitivity and specificity than TML models; however, these differences were not statistically significant (LRT χ^2 = 0.30, P = 0.86), indicating that algorithm type was not a significant source of heterogeneity. To explore the potential sources of heterogeneity and compare the effects of different factors on the diagnostic efficacy of models, this study conducted subgroup analyses on TML and DL based on dimensions such as sample size, validation strategy, Speech task type, language, Deep Learning Architecture, and diagnostic scale. Between-group differences were not statistically significant (all P for interaction > 0.05). The main findings are summarized in Table 1.

Sensitivity analyses

To assess the impact of individual studies on pooled effect sizes, a leave-one-out sensitivity analysis was performed for TML and DL models. For the 9 TML studies (11 models), excluding small-sample studies (Hashim et al. 2022a, sample size = 90) increased the pooled specificity from 0.83 to 0.84, while sensitivity and I^2 remained unchanged. In the 16 DL studies (20 models), excluding the extreme large-sample study (Ishimaru et al. 2023, sample size = 663978) yielded pooled sensitivity and specificity of 0.82 and 0.85, respectively, with no significant shifts, indicating that large-sample studies did not dominate the pooled effect sizes. Overall, the pooled sensitivity and specificity of both model types showed no substantial fluctuations after excluding any single study, confirming the robust stability of the meta-analysis results. The plots for the leave-one-out results of TML and DL can be found in Supplementary File (Figs. 2 and 3).

HSROC analyses showed no significant threshold effects for either TML (β = 0.23, 95% CI: 0.28–0.74, P = 0.39) or DL models (β = 0.13, 95% CI: 0.29–0.55, P = 0.54). Pooled sensitivity and specificity estimate from HSROC (TML: 0.82/0.83; DL: 0.83/0.86) were virtually identical to those from the bivariate model, confirming the robustness of our findings (Supplementary Tables S3–S4).

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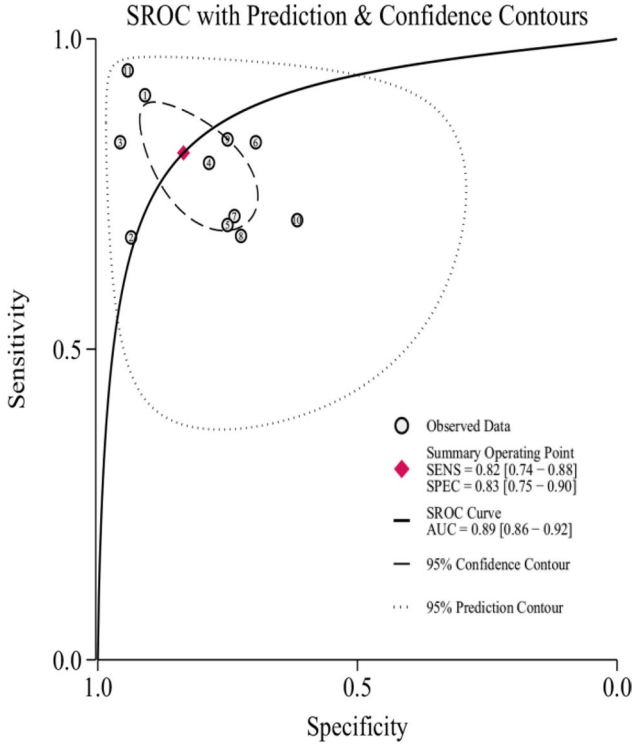


Fig. 4 SROC with prediction and confidence contours of TML

Publication bias analysis

Publication bias was assessed using Deeks' funnel plot asymmetry test, which is recommended for diagnostic test accuracy meta-analyses because conventional methods such as Egger's or Begg's tests are not appropriate when effect sizes are expressed as diagnostic odds ratios (DOR). The Deeks' test showed no significant asymmetry for TML models ($P = 0.79$). Although initial analysis indicated asymmetry for DL models ($P = 0.01$), this was driven by one extreme large sample study (Ishimaru et al., 2023). After excluding this outlier, the test became non-significant ($P = 0.98$), suggesting the absence of small-study effects. Sensitivity analyses confirmed that removing this study did not materially change pooled sensitivity or specificity.

However, given the limited number of included studies, the statistical power of this test may be insufficient. Therefore, these results should be interpreted with caution.

Discussion

Summary of the main findings

We conducted a review of the performance of DL and TML in depression detection using speech samples. This meta-analysis (9 TML studies and 16 DL studies, 706,945 speech samples) shows that both TML and DL models exhibit favorable performance in leveraging speech features for depression detection. Specifically, DL demonstrated a pooled sensitivity similar to that of TML (0.83 vs. 0.82), but with higher pooled specificity (0.86 vs. 0.83) and AUC (0.91 vs. 0.89). Subgroup analysis reveals that DL models trained on datasets with a sample size ≥ 500 achieve an AUC of 0.93. In the subgroup with a sample size > 100, the specificity of TML models increases from 0.76 to 0.87. Further subgroup analysis uncovers language-related performance disparities: the English group shows a higher specificity (0.90) than the Chinese group (0.81). Additionally, the non - DSM group enhances the specificity of TML models (0.71 in the DSM group vs. 0.89 in non - DSM scale-based groups).

Comparison with other review

Our findings are consistent with Liu et al. (2024) in terms of pooled specificity (0.85) and sensitivity (0.82). A key discrepancy emerges, however, in the performance comparison between DL and TML models. While Liu et al. reported higher accuracy for manually engineered features (0.89 vs. 0.82 for end-to-end architectures), our meta-analysis indicates that DL models achieved marginally superior performance (AUC: 0.91 vs. 0.89 for TML). We hypothesize that this discrepancy stems from critical differences in the included studies and methodological evolution, primarily concerning two factors.

First, our review included more recent studies utilizing larger and acoustically more diverse speech corpora (e.g., sample sizes ≥ 500). This scale of data allows DL architectures to fully leverage their capacity for hierarchical feature learning, capturing subtle, non-linear temporal dynamics (e.g., complex pitch fluctuation) that might be missed by static, manual features in TML(34,35,39). In contrast, the smaller, less diverse datasets predominant in earlier reviews might have constrained DL's generalization, favoring more robust TML approaches.

Second, many DL studies in our analysis employed techniques like data augmentation and transfer learning, which enhance robustness and were less common in earlier works [40]. The end-to-end training paradigm also optimizes the synergy between feature extraction and classification.

Therefore, the superior performance of DL in our analysis likely reflects a combination of larger, more representative datasets and more advanced modeling techniques, highlighting the context-dependent nature of model efficacy.

Subgroup and sensitivity analyses

Sample size

DL models (e.g., CNN, Transformer) automate the extraction of complex, high-dimensional nonlinear features from raw speech data, which enables superior performance (AUC: 0.93) when trained on large-scale datasets ($n \geq 500$). However, this capacity for automated

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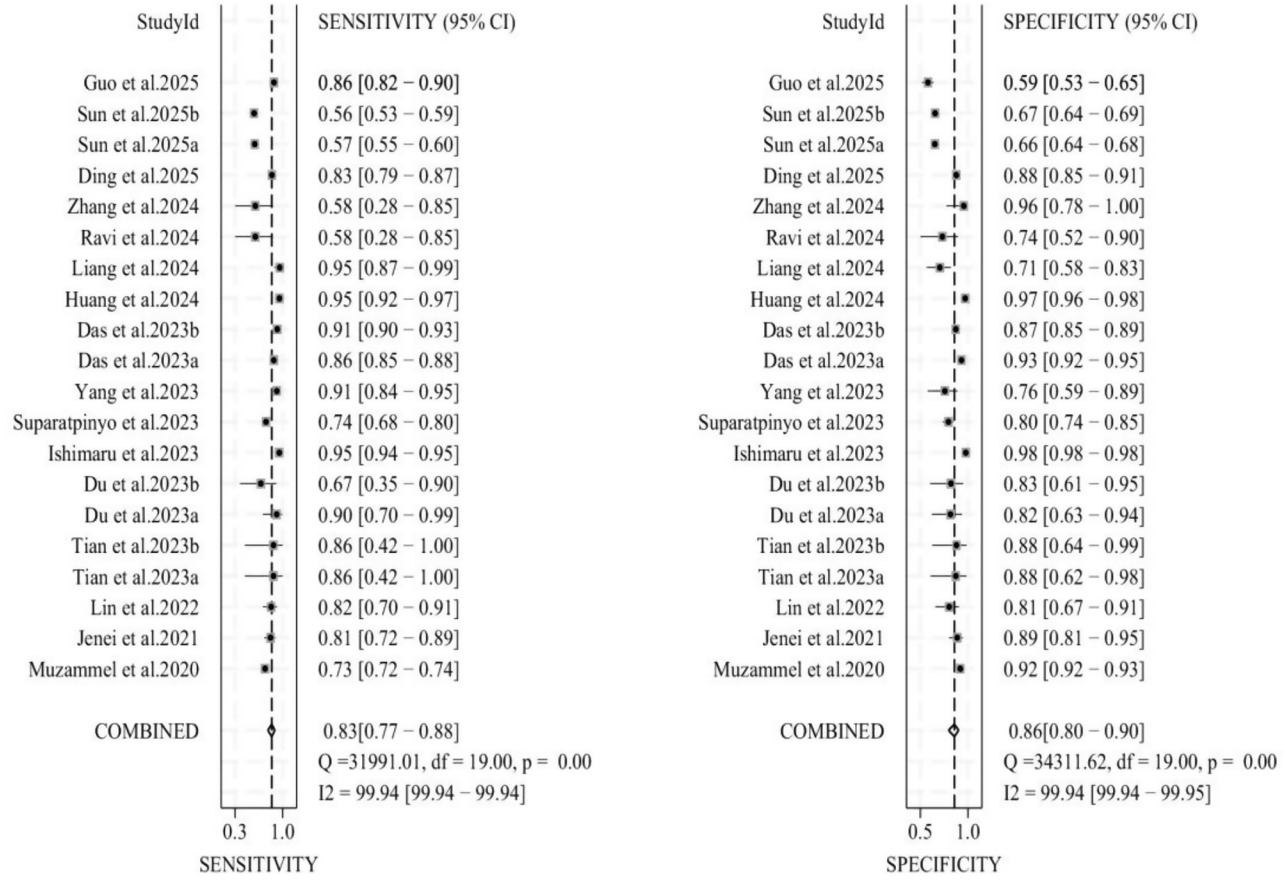


Fig. 5 The pooled sensitivity and specificity forest plots of DL

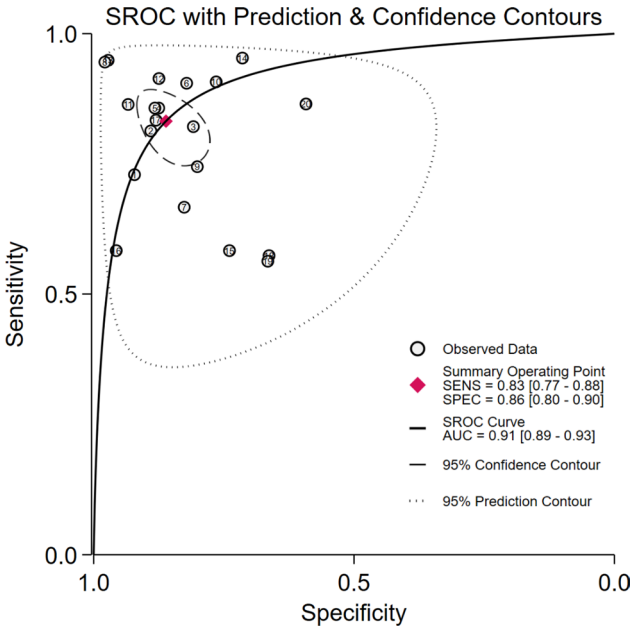


Fig. 6 SROC with prediction and confidence contours of DL

feature learning is not only highly data-dependent but also incurs substantial computational costs, including significant GPU memory requirements and prolonged training times [31]. In contrast, TML models, which rely on manually engineered features (e.g., MFCCs), achieve robust and data-efficient performance. As evidenced in our subgroup analysis, TML's specificity increased from 0.76 to 0.87 with larger samples, demonstrating its ability to refine statistical estimations without the prohibitive computational overhead of DL. This fundamental dichotomy underscores that model selection should not be guided by peak accuracy alone but by a pragmatic trade-off between performance and computational feasibility.

Language specificity

In our language subgroup analysis, the English group exhibited a higher specificity (0.90) than the Chinese group (0.81). This discrepancy likely stems from typological differences between languages. As a stress-timed language, English exhibits acoustically more discernible prosodic attenuation (e.g., reduced pitch variability, slower speech rate) associated with depression [41]. In contrast, tonal languages like Chinese have inherent pitch variations (tones) that may obscure or confound

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pathological prosodic changes, complicating model differentiation [39]. Beyond linguistic structure, cultural norms shape how depressive symptoms are expressed in speech. Differences in metaphor use, indirect expression, or focus on somatic complaints can alter prosodic cues such as pitch, intensity, and pauses. These culturally mediated expression styles can affect the acoustic patterns captured by detection models, affecting model performance [42, 43].

Validation strategy

The choice of validation strategy can also influence the reported performance of TML models. Grouped validation (e.g., LOOCV, k-fold CV) yielded considerably higher and likely more reliable AUC (0.93) compared to independent test set validation (0.77), especially given the modest sample sizes in TML studies. This underscores the vulnerability of TML performance estimates to evaluation methodology in small-sample scenarios and emphasizes the critical need for rigorous, repeated validation approaches to avoid over- or under-estimation of true model capability. In contrast, DL model performance showed less dependence on the specific validation approach (Grouped AUC: 0.92; Non-grouped AUC: 0.91), likely due to their greater inherent stability with larger datasets, provided the test set itself is truly independent and representative.

Diagnostic scales

We also found that when TML models used non-DSM scales (such as PHQ-9), the specificity reached 0.89, which was higher than 0.71 when using DSM scales. The DSM requires meeting specific symptom thresholds and duration (≥ 2 weeks), potentially excluding individuals with subclinical depression or those in very early stages whose speech may already exhibit subtle changes detectable by ML models [44]. In contrast, continuous scales like the PHQ-9 can capture a wider spectrum of symptom severity, aligning better with the potentially more sensitive detection capability of ML models based on speech68 [45]. This highlights the importance of aligning the ML task (e.g., detecting current symptom severity vs. strict DSM diagnosis) with the chosen reference standard for model training and evaluation.

Speech task type

Subgroup analyses showed better performance for interview-based than reading tasks in both TML and DL models. Spontaneous speech captures pauses, hesitations, and pitch changes linked to depressive psychomotor and affective symptoms, whereas reading constrains variability and suppresses diagnostic cues [46]. Interviews also engage cognitive and social-emotional processes impaired in depression, further enhancing signal [43, 47]. The modest but consistent inferiority of reading tasks suggests that models trained solely on scripted speech risk limited generalizability, underscoring the importance of spontaneous or naturalistic speech for future model development.

Deep learning architectures

Our subgroup analysis revealed a finding: traditional CNNs demonstrated superior performance compared to advanced CNN architectures, with higher sensitivity 0.86 (95% CI: 0.80–0.90) vs. 0.81 (95% CI:0.70–0.89), specificity 0.90 (95% CI:0.84–0.94) vs. 0.82 (95% CI:0.71–0.90), and AUC 0.94 (95% CI:0.91–0.96) vs. 0.89 (95% CI:0.85–0.91). These results suggest that accuracy is not driven by complexity alone. Advanced CNNs require larger datasets and may be overfit in small samples, while traditional CNNs benefit from established preprocessing (e.g., MFCCs, spectrograms) that enhances generalizability [48–50]. Inconsistent validation further undermines advanced CNN performance [51, 52]. Thus, architectural innovation does not guarantee diagnostic gains; aligning model complexity with data sufficiency and applying transfer learning or multi-corpus training may be essential.

Limitations

Several limitations should be acknowledged when interpreting the findings of this meta-analysis.

First, although both TML and DL models demonstrated good diagnostic accuracy, substantial heterogeneity remained across studies (I^2 for sensitivity and specificity generally exceeding 80%), indicating methodological and contextual variability rather than random error. This heterogeneity could be partly quantified and traced to identifiable sources. For TML models based on standardized rating scales such as the PHQ-9 rather than DSM or ICD interviews, I^2 was reduced by approximately 25% points. These findings indicate that both sample size and diagnostic reference standards materially influenced model stability.

Second, most of the included studies adopted cross-sectional designs, which restrict causal inference and prevent examination of longitudinal changes in speech characteristics or treatment-response trajectories. Future longitudinal or repeated-measure studies are needed to verify whether speech-derived biomarkers can sensitively track the course of depression or predict relapse.

Third, the linguistic distribution of included datasets was highly imbalanced, with approximately 68% based on English-language corpora. This not only limits its generalizability to tonal or morphologically complex languages but also reflects potential selection bias introduced by database choices, since most indexed databases such as PubMed, IEEE, and Scopus primarily

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archive English-language research. Consequently, non-English studies, especially those from East Asian and Middle Eastern populations, may have been underrepresented, leading to an overestimation of pooled accuracy in English corpora. In addition to linguistic differences, cultural norms also shape how depressive symptoms are expressed in speech. Variations in metaphor use, indirect expression, or emphasis on somatic versus emotional complaints may alter prosodic and acoustic patterns, potentially affecting model accuracy. This highlights the need for culturally sensitive validation to ensure generalizability across diverse clinical settings.

Fourth, incomplete reporting limited the precision of pooled estimates. Nearly one-third of potentially eligible studies did not provide sufficient data to reconstruct 2 × 2 tables, such as missing true positive, false positive, true negative, or false negative counts. This precluded calculation of standard diagnostic indices and may have introduced reporting bias favoring positive findings.

Clinical implementation challenges

To address these issues, future research should focus on (1) standardizing preprocessing and feature extraction procedures, including consistent frame segmentation and noise-reduction settings; (2) adopting uniform diagnostic reference standards and transparent data-splitting strategies; (3) performing meta-regression analyses to quantify how sample size, diagnostic criteria, and language explain between-study variance; (4) broadening database coverage to incorporate non-English studies; and (5) fully reporting diagnostic confusion matrices and confidence intervals to support reproducibility and evidence synthesis.

The voiceprints and other identity information contained in voice data are highly sensitive and are strictly regulated by laws and regulations. We can develop technologies such as federated learning and differential privacy. Federated learning avoids the transmission of original voice data through distributed training, while differential privacy provides mathematically provable privacy guarantees by injecting noise [53]. Federated learning combined with Differential Privacy (DP) can train voiceprint models to predict speakers' acoustic features while preventing the leakage of identity information, achieving "data utility without exposing data" [54].

However, the feasibility of these methods in speech-based depression detection remains limited. Speech data are inherently heterogeneous due to variations in accent, background noise, and recording devices, which can hinder model convergence in FL settings. Moreover, gradients exchanged during FL may still leak sensitive attributes, and excessive DP noise can degrade the fine-grained acoustic cues (e.g., prosody, pauses) that are essential for depression detection. Current applications in neighboring tasks such as speaker verification demonstrate that FL + DP is technically feasible, but further research is needed to balance privacy protection and diagnostic performance in clinical speech-based depression models.

Clinical implementation strategy

Based on this meta-analysis, we propose a hierarchical diagnostic framework to optimize resource allocation across clinical settings.

Primary Care Screening (TML Models): TML models (AUC = 0.89) offer moderate accuracy, low computational cost, and good interpretability, making them ideal for initial screening in resource-constrained settings such as primary care, mobile apps, and telehealth. Depression screening in primary care is cost-effective: one-time adolescent screening costs $44,483/QALY, annual screening $66,822/QALY [55], and adult annual screening $11,134–34,065/QALY [56]. Collaborative care after screening yields $9,592–14,306/QALY and over 100 depression-free days [57, 58]. Automated TML-based speech screening could further improve cost-effectiveness by reducing staff time and per-case costs while enabling continuous passive monitoring through smartphone applications that capture naturalistic speech during daily activities.

Secondary/Tertiary Care Confirmation (DL Models): For individuals flagged by screening, DL models (AUC = 0.91) can be deployed in psychiatric clinics or specialized services where computational resources are available. Their capacity to capture complex speech dynamics supports higher-confidence diagnostic confirmation, differential diagnosis, and longitudinal treatment monitoring.

Developing explainable AI (XAI) methods is essential to address the "black box" problem and foster clinician trust. Techniques such as SHAP, LIME, or attention visualization can reveal which acoustic cues (e.g., pitch, speech rate, pauses) drive model predictions, thereby improving transparency, aligning results with clinical knowledge, and supporting adoption in real-world practice.

Importantly, the choice of algorithm should also be aligned with the clinical setting. In primary care or community health centers, where computational resources are limited, TML models may offer a pragmatic solution for preliminary screening or frequent monitoring. Conversely, in psychiatric hospitals or specialized mental health services, DL models can leverage greater computational capacity and richer datasets to provide more accurate diagnostic support and longitudinal monitoring. Tailoring model deployment to clinical context will maximize feasibility and clinical utility.

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Future research directions

Based on this review's limitations, several future research directions are essential to advance the field:

Improve methodological rigor Future studies must prioritize rigorous design. This includes adopting strict, representative patient selection criteria and ensuring a clear separation between model development and validation to prevent data leakage.

Establish standardized protocols To reduce heterogeneity and allow for meaningful cross-study comparisons, standardized protocols for speech feature extraction and model evaluation must be established.

Adopt longitudinal designs Longitudinal research is essential to capture the dynamic evolution of speech features in relation to depression onset, progression, and treatment response, offering insights into potential causal relationships.

Integrating real-time monitoring Future studies should investigate how speech-based depression detection models can be effectively embedded in smartphone applications and wearable devices to enable continuous, real-world monitoring. Such integration would allow early detection of symptom deterioration, dynamic assessment of treatment response, and relapse prevention through timely alerts. Additionally, research should address challenges related to data heterogeneity, privacy, and integration with clinical workflows. This direction holds potential to transform episodic diagnostic assessments into continuous mental health monitoring systems.

Enhance interpretability (XAI) To address the "black box" challenge, future work should focus on developing XAI methods. This is key to fostering clinical trust and adoption.

Broaden evidence synthesis Finally, future systematic reviews must expand their scope by incorporating grey literature and unpublished data to mitigate publication bias and provide a more comprehensive picture of model performance.

Conclusions

In this systematic review and meta-analysis, we evaluated the diagnostic accuracy of traditional machine learning and deep learning models for speech-based depression detection. Both approaches demonstrated promising performance, with deep learning showing marginal advantages. Subgroup analyses indicated that factors such as sample size, validation strategy, language, and diagnostic reference standards influenced outcomes. Despite these encouraging findings, substantial heterogeneity, limited linguistic diversity, and the predominance of cross-sectional designs limit the strength of the evidence. To enhance clinical applicability and generalizability, future studies should adopt more rigorous designs, include multilingual datasets, and follow standardized reporting frameworks such as PRISMA-Abstract and STARD-AI to ensure transparency and reproducibility.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12888-025-07628-z.

Supplementary Material 1

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Author contributions

Wei Lu: Conceptualization, Methodology, Software, Visualization, Writing – original draft. Tang Xiaowei: Methodology, Data curation, Visualization. Dongmei Wu: Conceptualization, Methodology, Supervision, Writing - reviewing and editing, Funding acquisition. Man Wei: Methodology, Data curation, Visualization. Chuan Huang: Methodology, Data curation, Visualization. Chengxin Bai: Methodology, Data curation, Visualization. Xuqing Fan: Methodology, Data curation, Visualization.

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Data availability

All data analyzed during this study are included in this published article and the original studies' publication.

Declarations

Ethics, consent to participate, and consent to publish

Not applicable.

Competing interests

The authors declare no competing interests.

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